

Computing Practical - Programming Cheatsheet

2.3.4 with Statements

Format with open('name of file', 'w') as f: → write()
with open('name of file', 'r') as f: → read(), readlines()

Unpacking read() / readlines()

1. read()
data = f.read()
data = data.split("\n")
data = data[:-1]

2. readlines()
data_lines = f.readlines()
newdata = []
for data in data_lines:
newdata.append(data.strip("\n"))

2.3.7 Integer & Floats

1. Rounding

round(x)
math.floor(x) → round down
math.ceil(x) → round up
math.trunc(x)

2. Square Roots

math.sqrt(x)

3. Ranges

range(x) → 0 to x-1
range(x, y) → x to y-1
range(x, y, z) → x to y-1,
increment by z

4. Random Numbers

random.randint(x, y)
→ random int between x and y
(inclusive)
random.random()
→ random float between
0 (inclusive) and 1 (exclusive)

2.3.8 Strings

1. Indexing & Slicing

str[a]
str[a:b], str[a:b:c]

2. ASCII

chr(int), ord(str)

3. Find

str.find(x)
str.find(x, a, b)

4. Membership

substr in str

5. Split

str.split() → separate by space
str.split(x) (returns list)

6. Upper & Lower

str.upper()
str.lower()

7. Methods returning bool

str.isalpha()
str.isalnum()
str.islower()
str.isupper()
str.isdigit()
str.isspace()
str.startswith(x)
str.endswith(x)

2.3.9 Lists

1. Indexing & Slicing

list[a]
list[a:b], list[a:b:c]

2. Membership

value in list

3. sum, max, min

sum(list)
max(list)
min(list)

4. In-place Methods (do not assign)

list.append(value)
list.insert(index, value)
list.pop(index)
list.remove(value)
list.clear()
list.sort()
list.reverse()

2.3.10 DICTIONARIES

1. Insertion
dict[new_key] = value
2. Lookup
dict[key] or dict.get(key)
3. Query
key in dict
value in dict.values()
4. Deletion
del dict[key]
del dict

2.3.12 LOOPS

1. For
for item in iterable:
* dict → item refers to key
for i in range(x, y):
→ use only if index matters
2. While
while (condition):
→ use when unknown number of loops
3. Break/continue
break
continue

2.4.6 Input Validation

1. Interactive Input
(Ask for input again)
while True:
 user_input = input(" ")
 if (conditions for valid):
 break
 else:
 print("Invalid Input")
2. Non-Interactive Input
(Exit Program)
import sys
if not (conditions for valid):
 sys.exit()

2.5.1 Algorithms

1. min() / max()
numbers = [list]
smallest = numbers[0]
for number in numbers:
 if number < smallest:
 smallest = number
largest = numbers[0]
for number in numbers:
 if number > largest:
 largest = number
2. Sum and Average
numbers = [list]
total = 0
for number in numbers:
 total += number
average = total / len(numbers)
3. Index() and Find()
mylist = [list]
mystr = ""
target = (target value)
for i in range(len(mylist/mystr)):
 if mylist/mystr[i] == target:
 print(i)
4. Extracting items/values
mylist = [list]
mystr = ""
result = [] or ""
for value in mylist/mystr:
 if (condition):
 result.append(value) → list
 result += value → str
5. Split()
mystr = ""
words = []
current = ""
delimiter = (delimiter)
for char in mystr:
 if char == delimiter:
 words.append(current)
 current = ""
 else:
 current += letter
words.append(current)